Self-Supervised Pneumonia Detection from Chest X-Rays via I-JEPA:

A Latent-Space Predictive Approach with Label-Efficient Learning

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Abstract. Pneumonia causes over 2.2 million deaths annually, including 502,000 children under five, yet accurate chest X-ray (CXR) interpretation remains inaccessible in resource-limited settings due to critical radiologist shortages. Existing deep learning diagnostic systems are further constrained by the scarcity of expert-annotated medical images, motivating the need for annotation-efficient learning approaches. This study applies I-JEPA (Image-based Joint-Embedding Predictive Architecture) to binary pneumonia classification by pretraining a ViT-Small/16 encoder on 50,000 unlabeled NIH ChestX-ray14 images via random patch masking, then fine-tuning on the RSNA Pneumonia Detection Dataset. Unlike pixel-reconstruction methods, I-JEPA predicts masked-region latent representations, compelling the encoder to acquire high-level anatomical features directly relevant to clinical diagnosis. Full fine-tuning achieves AUC = 0.8297 and Recall = 0.7738, consistently outperforming ImageNet-pretrained baselines in recall across all decision thresholds; at 1% labeled data (187 images), I-JEPA attains AUC = 0.8041 ± 0.0002 surpassing ResNet50 and ViT-Small/16 while exhibiting 143-fold lower performance variance. These findings establish I-JEPA as a practical, annotation-efficient framework for automated pneumonia screening. A key secondary finding is that CLAHE preprocessing uniformly degrades performance across all models (mean ΔAUC = −0.047, range: −0.015 to −0.069), revealing a previously undocumented interaction between SSL pretraining and preprocessing pipelines: preprocessing must remain consistent between pretraining and fine-tuning to preserve learned domain-specific representations. This result provides a critical design guideline for SSL-based CXR pipelines.

Keywords: *Self-Supervised Learning; Chest X-Ray; Pneumonia Classification; I-JEPA; Label Efficiency*

1 Introduction

Pneumonia remains one of the most significant infectious causes of morbidity and mortality worldwide, disproportionately affecting young children and older adults.

According to the World Health Organization, approximately 2.2 million deaths were attributed to pneumonia in 2021, including more than 502,000 children under five years of age (WHO, 2024). Early and accurate diagnosis is critical for improving patient outcomes, as timely treatment can substantially reduce disease severity and mortality. Consequently, developing reliable and accessible diagnostic tools remains an important public health priority, particularly in low-resource healthcare settings.

Chest X-ray (CXR) imaging is the most widely used modality for pneumonia diagnosis due to its low cost, rapid acquisition, and broad availability across different levels of healthcare infrastructure. However, accurate interpretation of CXR images requires specialized radiological expertise that is not uniformly available. In many developing countries, including Vietnam, district hospitals and primary healthcare centers often lack permanently stationed radiologists, while the increasing volume of medical imaging examinations places considerable workload pressure on available clinicians. These challenges can contribute to delayed diagnoses and increased rates of interpretation errors, highlighting the need for automated decision-support systems capable of assisting healthcare professionals in routine clinical practice. Recent advances in deep learning have demonstrated remarkable success in medical image analysis. Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs) have achieved competitive performance on various chest radiography benchmarks, including pneumonia and thoracic disease classification tasks (Wang et al., 2017; Dosovitskiy et al., 2021). Nevertheless, most existing approaches rely heavily on supervised learning and therefore require large quantities of expert-annotated data — a major limitation in medical imaging where annotation is expensive, time-consuming, and dependent on highly trained specialists.

To address these limitations, self-supervised learning (SSL) has emerged as a promising paradigm by leveraging large collections of unlabeled medical images to learn informative visual representations prior to downstream fine-tuning. Previous studies have shown that domain-specific SSL pretraining can outperform conventional ImageNet-supervised initialization, particularly when labeled data are scarce (Azizi et al., 2021). Among recent SSL approaches, the Image-based Joint-Embedding Predictive Architecture (I-JEPA) proposed by Meta AI (Assran et al., 2023) represents a notable advancement. Unlike reconstruction-based methods such as Masked Autoencoders (MAE) (He et al., 2022), which learn by recovering masked pixels, or contrastive methods such as SimCLR (Chen et al., 2020) and BYOL (Grill et al., 2020), which depend on carefully designed data augmentations, I-JEPA learns to predict latent representations of masked image regions directly in embedding space. This predictive objective encourages the model to capture high-level semantic and structural information, making it particularly attractive for medical imaging applications where clinically relevant findings are often defined by anatomical patterns rather than low-level image appearance. Despite these theoretical advantages, the effectiveness of I-

JEPA for pneumonia classification from chest X-ray images remains largely unexplored.

To address these gaps, this study evaluates an I-JEPA-based framework for binary pneumonia classification using chest X-ray images. Specifically, three research questions are investigated: (RQ1) whether I-JEPA pretraining on unlabeled chest X-rays can improve classification performance compared with ImageNet-supervised pretraining; (RQ2) whether I-JEPA exhibits superior label efficiency across different labeled-data regimes ranging from 1% to 100% of the training set; and (RQ3) whether the regions emphasized by the fine-tuned I-JEPA model correspond to clinically relevant pneumonia lesions annotated by radiologists. The main contributions of this work are threefold. First, a systematic evaluation of I-JEPA with random patch masking for binary pneumonia classification on the RSNA Pneumonia Detection benchmark is presented. Second, a comprehensive label-efficiency analysis comparing I-JEPA against representative CNN and Transformer baselines across six labeled-data fractions (1%–100%) is performed. Third, an explainability study using Grad-CAM (Selvaraju et al., 2020) and Attention Rollout methods is conducted, quantitatively evaluating the alignment between model attention maps and radiologist-provided lesion annotations.

2 Related Work

2.1 Chest X-Ray Classification and Supervised Learning

Wang et al. (2017) established the foundational benchmark for large-scale CXR classification through the NIH ChestX-ray14 dataset (>112,000 images, 14 NLP-derived pathology labels). Despite enabling early CNN-based classification at scale, the NLP-derived labels carry inherent noise (estimated accuracy ≈ 90%), and the pneumonia-specific label constitutes only approximately 1.3% of total samples. The RSNA Pneumonia Detection Challenge (RSNA, 2018) subsequently provided approximately 30,000 expert-verified binary labels with radiologist bounding boxes, establishing the de facto standard for pneumonia detection benchmarking. In the Vietnamese clinical context, Khiem et al. (2026) demonstrated the feasibility of supervised deep learning for pulmonary disease classification, highlighting both the growing domestic research momentum and the persistent challenge of labeled data scarcity at the primary care level.

2.2 Self-Supervised Learning for Visual Representation Learning

He et al. (2020) introduced MoCo, employing a momentum-updated encoder and memory queue to learn representations by contrasting augmented views. Chen et al. (2020) proposed SimCLR, demonstrating that strong augmentation and large batch sizes yield competitive ImageNet representations. However, both methods introduce risks of false negatives in the medical domain two CXR images sharing the same pathological category may be incorrectly pushed apart in representation space and impose prohibitive GPU memory requirements. Grill et al. (2020) proposed BYOL, eliminating negative pairs through an EMA-updated target network. BYOL introduced the EMA mechanism later inherited by I-JEPA, though its reliance on augmentation-induced invariance may suppress clinically meaningful variation in CXR images. In parallel, He et al. (2022) introduced MAE (Masked Autoencoders), shifting the SSL objective to within-image reconstruction: 75–80% of image patches are randomly masked, and the model reconstructs masked pixel values. However, pixel-level reconstruction forces the encoder to allocate representational capacity toward low-level surface details rather than the high-level semantic features that drive clinical interpretation of CXR images.

Within the medical imaging domain, Azizi et al. (2021) provided the first systematic empirical evidence that in-domain SSL pretraining on unlabeled medical images consistently outperforms ImageNet-pretrained baselines in the low-label regime. Using a SimCLR-based framework (MICLe), the study showed that SSL models with as little as 1% labeled data can approach fully supervised baselines, establishing the core premise motivating the present study. More recently, Assran et al. (2023) proposed I-JEPA, resolving the limitations of prior SSL approaches by predicting in latent space. Several domain adaptations confirm the flexibility of the masking strategy: Fei et al. (2024) (A-JEPA) and Tuncay et al. (2025) (Audio-JEPA) demonstrate random masking achieves state-of-the-art on audio spectrograms; Xu et al. (2025) (CrossJEPA) shows JEPA remains effective without any masking; Zhao et al. (2024) (Brain-JEPA) applies spatiotemporal masking to fMRI data; and Alis et al. (2026) (RadJEPA) applies JEPA directly to unlabeled CXR images with latent prediction objectives, achieving state-of-the-art on multiple radiographic tasks.

2.3 Research Gaps

Synthesizing the above, three specific gaps remain unaddressed: (1) No study has systematically evaluated I-JEPA pretraining for binary pneumonia classification on the RSNA benchmark with direct comparison against CNN and Transformer baselines sharing an identical encoder architecture. (2) No label-efficiency analysis has been conducted for I-JEPA versus supervised baselines on CXR data across a granular range of labeled fractions. (3) No quantitative explainability evaluation has been performed for I-JEPA on CXR data using radiologist bounding box annotations as ground truth. The present study directly addresses all three gaps.

3 Materials and Methods

3.1 Datasets

Two publicly available datasets are employed in this study, each with a clearly delineated role in the experimental pipeline, as summarized in Table 1.

Table 1. Datasets used in this study

Dataset	Role	Size	Labels Used
NIH ChestX-ray14 (Wang et al., 2017)	Self-supervised pretraining	50,000 images (subset)	None
RSNA Pneumonia Detection (RSNA, 2018)	Fine-tuning & evaluation	26,684 images	Binary expert labels + bounding boxes

NIH ChestX-ray14. Wang et al. (2017) released over 112,000 frontal CXR images with 14 NLP-derived pathology labels. In this study, 50,000 images are randomly sampled from the full corpus as an unlabeled pretraining source, using a pre-resized 224×224 version. The 14 pathology labels are explicitly excluded from all experimental conditions: label accuracy is estimated at approximately 90% for some classes (Wang et al., 2017), and the pneumonia-specific label (≈1.3% of total samples) is insufficient in quantity and reliability for specialized classifier training. Zero patient-level overlap with RSNA is confirmed via patient identifier comparison and the RSNA challenge documentation.

RSNA Pneumonia Detection Dataset. The Radiological Society of North America organized the Pneumonia Detection Challenge in 2018 (RSNA, 2018), providing approximately 30,000 frontal chest X-ray images with expert radiologist binary labels and lesion bounding boxes for positive cases. The dataset is partitioned into training, validation, and test subsets using stratified splitting (seed = 42). Class imbalance (approximately 1:3.44 positive-to-negative ratio) is addressed through the use of pos_weight within BCEWithLogitsLoss. The resulting data distribution is summarized in Table 2.

Table 2. RSNA dataset split distribution (stratified split, seed = 42)

Split	Total	Pneumonia	Normal	Positive Rate
Train	18,678	4,207 (22.5%)	14,471 (77.5%)	22.5%
Validation	4,003	902 (22.5%)	3,101 (77.5%)	22.5%
Test	4,003	902 (22.5%)	3,101 (77.5%)	22.5%
Total	**26,684**	**6,011**	**20,673**	**22.5%**

3.2 Preprocessing

A unified preprocessing pipeline is applied consistently across both datasets and all experimental conditions. RSNA images in DICOM format are read via pydicom, with 16-bit pixel arrays extracted and converted to 8-bit PNG. All grayscale images are converted to three-channel format by channel replication, required by the ViT-Small/16 architecture. Images are resized to 224×224 pixels and normalized using ImageNet statistics (mean = [0.485, 0.456, 0.406]; std = [0.229, 0.224, 0.225]).

Data augmentation is applied exclusively to the RSNA training split during fine-tuning; no augmentation is applied during I-JEPA pretraining or to the validation and test sets. Augmentation techniques are selected to be medically appropriate, as detailed in Table 3. Vertical flip, aggressive random crop, and strong color distortion are explicitly excluded to preserve clinically relevant spatial structure and intensity distributions.

Table 3. Data augmentation applied during fine-tuning (training set only).

Technique	Parameters	Rationale
Random horizontal flip	p = 0.5	CXR images exhibit approximate bilateral symmetry
Random rotation	±7°	Simulates minor patient positioning variation
Color jitter (brightness)	±0.2	Simulates inter-device intensity variation
Random affine (translate)	≤5%	Simulates minor spatial shift

CLAHE (Contrast Limited Adaptive Histogram Equalization; clip_limit = 2.0, tile_grid_size = 8×8) is evaluated as a separate ablation study rather than incorporated into the default workflow, following prior evidence that its impact on SSL-pretrained models can be highly dataset-dependent.

3.3 Proposed I-JEPA Framework

3.3.1 Architecture Overview

The proposed framework adapts I-JEPA (Assran et al., 2023) with random patch masking for chest X-ray pretraining. Three components operate jointly during pretraining.

Student Encoder (f_θ). A full ViT-Small/16 network (Dosovitskiy et al., 2021) with embedding dimension d = 384, 12 Transformer blocks, and 6 attention heads. The encoder receives all 196 patch tokens from a 14×14 grid (patch size 16×16 at 224×224 resolution) and produces 196 patch embeddings of dimension 384.

Target Encoder ($f\bar{\theta}$). An architectural copy of the Student Encoder whose parameters are updated exclusively through Exponential Moving Average (EMA):

$$\bar{\theta} \leftarrow m \cdot \bar{\theta} + (1 - m) \cdot \theta$$

Where the momentum m is linearly increased from 0.996 at epoch 1 to 0.999996 at epoch 50. This schedule ensures the Target Encoder provides increasingly stable prediction targets as training progresses.

Predictor (g ϕ). A narrow Transformer network with 4 layers, embedding dimension 256, and 4 attention heads. The Predictor receives 196 patch embeddings from the Student Encoder and produces predicted embeddings for the 147 randomly selected target positions (mask ratio = 0.75).

3.3.2 Random Patch Masking Strategy

At each training step, a binary mask $M \in \{0,1\}^{196}$ is sampled uniformly at random with 147 target positions (mask ratio = 0.75). The complementary 49 positions form the context set. The Student Encoder processes the full image, while the Predictor generates predictions only for the 147 masked target positions.

This random masking strategy departs from I-JEPA block masking for three considerations. First, implementation fidelity: correct block masking requires preventing the ViT encoder from attending to target patches during its forward pass, a non-trivial modification susceptible to silent implementation errors on the Kaggle T4 platform. Second, cross-domain evidence: Tuncay et al. (2025) and Fei et al. (2024) independently demonstrate that random masking achieves state-of-the-art performance when JEPA is transferred to new domains; Xu et al. (2025) further confirms JEPA remains effective without any masking; Alis et al. (2026), who directly apply JEPA to CXR data, further substantiate the viability of domain-adapted masking strategies in radiographic pretraining. Third, the high bilateral symmetry of frontal CXR images makes random masking well-suited for uniform anatomical coverage during pretraining.

3.3.3 *Training Objective*

The pretraining loss is the mean squared error (MSE) computed in latent space between the Predictor's output and the Target Encoder's embeddings at the masked positions:

$$\mathcal{L} = \frac{1}{|\mathcal{T}|} \sum_{i \in \mathcal{T}} \left| g_\phi\big(f_\theta(x)\big)[i] - f_{\bar{\theta}}(x)[i] \right|_2^2$$

where $\mathcal{T}$ denotes the set of 147 masked target positions, $g_\phi\big(f_\theta(x)\big)[i]$ is the Predictor's embedding at position i, and $f_{\bar{\theta}}(x)[i]$ is the Target Encoder's embedding at the same position. No pixel reconstruction is performed at any stage. The Student Encoder and

Predictor are updated via backpropagation through $\mathcal{L}$; the Target Encoder receives no gradient signal.

3.3.4 *Fine-Tuning Strategies*

Following self-supervised pretraining, the Student Encoder is augmented with a binary classification head (Linear(384, 1)) and fine-tuned under five experimental configurations: Linear Probing (LP), Partial Fine-Tuning (Partial FT), and Full Fine-Tuning (Full FT), each in two optimization variants (v1: fixed learning rates; v2: Layer-wise Learning Rate Decay, LLRD). Complete hyperparameter settings are summarized in Table 4.

Table 4. Fine-tuning hyperparameter configurations

Strategy	Hyperparameter	Config v1	Config v2
Linear Probe	Head LR	1×10^{-3}	1×10^{-3}
Linear Probe	Epochs / Patience	15 / 5	15 / 5
Partial FT	Head LR	1×10^{-4}	3×10^{-5}
Partial FT	Encoder LR	1×10^{-5} (flat)	5×10^{-6} + LLRD
Partial FT	Unfrozen blocks	Last 4	Last 4
Partial FT	Epochs / Patience	20 / 6	25 / 8
Partial FT	Warmup ratio	10%	15%
Full FT	Head LR	5×10^{-5}	3×10^{-5}
Full FT	Encoder LR	5×10^{-6} (flat)	1×10^{-6} + LLRD
Full FT	Layer-wise decay	None	0.65
Full FT	Epochs / Patience	35 / 8 (resume)	35 / 8 (no resume)
Full FT	Warmup ratio	15%	20%
Shared	Classifier dropout	0.2	0.1
Shared	Gradient clipping	1.0	0.5
Shared	Weight decay	0.05	0.05

Shared	Loss function	BCEWithLogitsLoss + pos_weight	BCEWithLogitsLoss + pos_weight

3.4 Baseline Models

Two supervised baselines are employed to provide controlled comparisons. ResNet50 (He et al., 2016) is selected as a representative CNN architecture, initialized with IMAGENET1K_V2 pretrained weights (~25M parameters) from torchvision (Paszke et al., 2019), with the final fully connected layer replaced by a Linear(2048, 1) binary classification head. ViT-Small/16 (Dosovitskiy et al., 2021) serves as the Transformer-based baseline and shares the same backbone architecture as the proposed I-JEPA encoder, thereby isolating the effect of self-supervised pretraining from architectural differences. The ViT baseline (~22M parameters) is initialized using ImageNet-21k → ImageNet-1k pretrained weights via DeiT (Touvron et al., 2021), loaded from the timm library (Wightman, 2019) (checkpoint: vit_small_patch16_224), with a Linear(384, 1) classification head. Both baselines are trained using AdamW with BCEWithLogitsLoss (pos_weight = 3.44) on the same fixed RSNA split (seed = 42).

Rather than adopting published baseline results directly, both baselines are retrained from scratch under identical experimental conditions, including the same dataset split, preprocessing pipeline, loss function, and evaluation protocol as the proposed I-JEPA framework. This controlled design ensures that any observed performance differences are attributable solely to the pretraining method, eliminating confounding variables arising from dataset partitioning, class balancing strategies, or preprocessing heterogeneity across studies.

3.5 Experimental Setup

All experiments are implemented in PyTorch 2.x (Paszke et al., 2019) using the timm (Wightman, 2019) and pydicom libraries, executed on Kaggle Notebook environments equipped with two NVIDIA Tesla T4 GPUs (16 GB VRAM per GPU). During self-supervised pretraining, I-JEPA is trained on 50,000 unlabeled NIH ChestX-ray14 images for 50 epochs using AdamW optimization (peak LR = 1×10^{-4}), batch size 32 with gradient accumulation ×2 (effective batch size 64), mixed-precision FP16 training, and a masking ratio of 0.75. The EMA momentum is linearly increased from 0.996 to 0.999996 throughout training. The pretraining MSE loss decreases from 0.1185 at epoch 1 to 0.0041 at epoch 50, indicating stable convergence without mode collapse. Pretraining required approximately 4 hours; full fine-tuning required approximately 6 hours on the described hardware.

3.6 Evaluation Metrics

Classification performance is primarily evaluated using the Area Under the Receiver Operating Characteristic Curve (AUC-ROC) as the principal metric, due to class imbalance in the RSNA dataset. Additional metrics - Accuracy, Precision, Recall (Sensitivity), Specificity, and F1-score are complemented by confusion matrices for class-wise error analysis. Formally, let *TP*, *TN*, *FP*, and *FN* denote true positives, true negatives, false positives, and false negatives on the RSNA test set, with Pneumonia as the positive class. These metrics are defined as:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Precision = \frac{TP}{TP + FP}$$

$$Recall\ (Sensitivity) = \frac{TP}{TP + FN}$$

$$Specificity = \frac{TN}{TN + FP}$$

$$F1 - Score = \frac{2 \cdot TP}{2 \cdot TP + FP + FN}$$

The AUC-ROC is defined as the area under the Receiver Operating Characteristic curve, obtained by plotting the True Positive Rate against the False Positive Rate across all decision thresholds $\tau \in [0, 1]$:

$$\text{AUC-ROC} = \int_0^1 \text{TPR}\left(\text{FPR}^{-1}(t)\right) dt$$

where $\text{TPR}(\tau) = \text{Recall}(\tau)$ and $\text{FPR}(\tau) = FP(\tau) / (FP(\tau) + TN(\tau))$. In practice, AUC-ROC is estimated from the discrete ROC curve using the trapezoidal rule.

To support clinical threshold selection, performance is additionally reported at the F1-optimal threshold τ^* and at recall-targeted thresholds, where:

$$\tau^* = \arg\max_{\tau \in [0,1]} \text{F1-Score}(\tau)$$

Youden's J statistic is used to identify the operating point that balances sensitivity and specificity equally:

$$J = \text{Recall} + \text{Specificity} - 1 = \frac{TP}{TP + FN} + \frac{TN}{TN + FP} - 1$$

Label efficiency is assessed across six labeled-data fractions (1%–100%). Performance is reported at two decision thresholds: the default threshold (0.50) and the F1-optimal threshold determined on the validation set. For explainability evaluation, Pointing Game Accuracy measures whether the peak activation falls within the annotated lesion region, while Overlap Ratio quantifies the spatial correspondence between attention maps and expert annotations.

4 Results

4.1 Pretraining Convergence

The I-JEPA model was pretrained for 50 epochs on 50,000 unlabeled NIH ChestX-ray14 images. The MSE loss in latent space decreased monotonically from 0.1185 at epoch 1 to 0.0041 at epoch 50, representing a 96.5% reduction, with no loss increase or mode collapse throughout training. This smooth convergence profile confirms stable optimization under the EMA cosine schedule. The loss curve is presented in Figure 1.

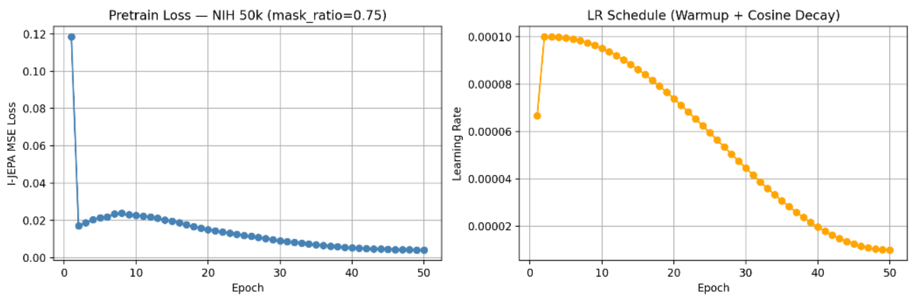

Figure 1. MSE latent-space loss during I-JEPA pretraining on NIH ChestX-ray14 (50 epochs). The loss decreases monotonically from 0.1185 to 0.0041 without mode collapse, indicating stable convergence.

4.2 Baseline Performance

Table 5 reports the performance of the two supervised baselines on the RSNA test set at both the default threshold (0.50) and the F1-optimal threshold.

Table 5. Baseline model performance on the RSNA test set (4,003 images; 902 positive cases).

Model	AUC	F1 (thr=0.50)	Recall (thr=0.50)	Specificity (thr=0.50)	F1 (best-thr)	Best Threshold
ResNet50 (He et al., 2016)	0.8862	0.6249	0.5421	0.9439	0.6508	0.37
ViT-Small/16 (Dosovitskiy et al., 2021)	0.8797	0.4169	0.2783	0.9836	0.6379	0.24

ResNet50 achieves the highest overall AUC (0.8862) across all experimental conditions. However, at the default threshold its Recall is only 0.5421, indicating that nearly 46% of true pneumonia cases would be missed in a clinical screening scenario. ViT-Small/16 pretrained on ImageNet yields a comparable AUC (0.8797) but exhibits a substantial domain-induced calibration shift: at threshold 0.50, Recall collapses to 0.2783. The optimal F1 threshold of 0.24 required to recover competitive ViT performance is a direct indicator of the domain gap between ImageNet natural images and chest radiographs.

4.3 I-JEPA Fine-Tuning Results

Table 6 reports the performance of all I-JEPA fine-tuning configurations on the RSNA test set.

Table 6. I-JEPA fine-tuning results on the RSNA test set. LLRD = Layer-wise Learning Rate Decay.

Configuration	AUC	F1 (thr=0.50)	Recall (thr=0.50)	Specificity (thr=0.50)	F1 (best-thr)	Recall (best-thr)	Best Thr.
Linear Probe	0.7737	0.5324	0.6918	0.7362	0.5344	0.7062	0.49
Partial FT v1	0.8003	0.5550	0.7417	0.7291	0.5630	0.6907	0.56
Partial FT v2 (LLRD)	0.8011	0.5525	0.7029	0.7552	0.5596	0.6608	0.55
Full FT v1	**0.8297**	0.5752	**0.7738**	0.7333	0.5921	0.6896	0.60
Full FT v2 (LLRD)	0.8041	0.5552	0.7195	0.7462	0.5661	0.6863	0.55

Three key observations emerge. First, classification performance improves monotonically as more encoder layers are unfrozen (AUC: 0.7737 → 0.80 → 0.8297), confirming that task-specific adaptation is necessary while the non-trivial Linear Probe AUC validates the quality of pretrained representations. Second, Full FT v1 achieves Recall = 0.7738 at threshold 0.50 approximately 2.77× higher than ViT-ImageNet (0.2783) and +0.232 over ResNet50 (0.5421). Since both I-JEPA Full FT v1 and ViT-ImageNet share the same ViT-Small/16 encoder architecture, this advantage is directly attributable to self-supervised in-domain pretraining. Third, Full FT v2 with LLRD underperforms Full FT v1, an outcome that may be attributed to the absence of sequential fine-tuning initialization available to v1.

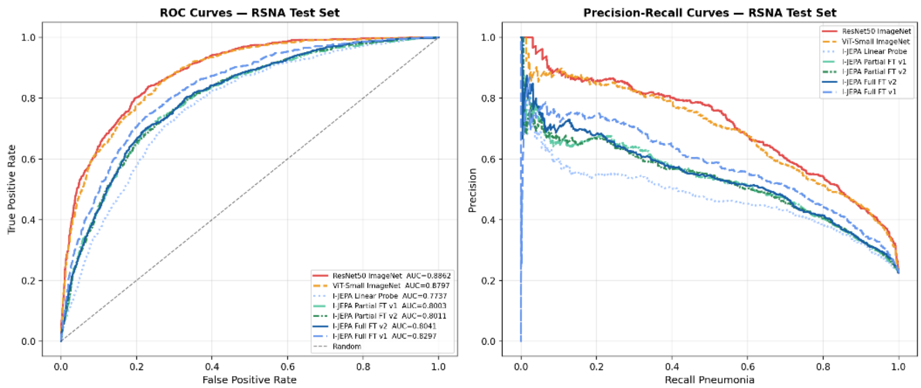

Figure 2. ROC curves (left) and Precision-Recall curves (right) for all models on the RSNA test set.

4.4 Comparative Summary

Table 7. Comparative summary across all experimental configurations, sorted by AUC.

Model	Group	AUC	Recall (thr=0.50)	F1 (best-thr)	Best Thr.
ResNet50 ImageNet	CNN Baseline	**0.8862**	0.5421	0.6508	0.37
ViT-Small/16 ImageNet	ViT Baseline	0.8797	0.2783	0.6379	0.24
I-JEPA Full FT v1	Proposed	0.8297	**0.7738**	0.5921	0.60
I-JEPA Partial FT v2	Proposed	0.8011	0.7029	0.5596	0.55

I-JEPA Partial FT v1	Proposed	0.8003	0.7417	0.5630	0.56
I-JEPA Linear Probe	Proposed	0.7737	0.6918	0.5344	0.49

With respect to RQ1, the architecturally controlled comparison between I-JEPA Full FT v1 and ViT-Small/16 ImageNet which share the same ViT-Small/16 encoder reveals a Recall advantage of +0.495 at threshold 0.50 for I-JEPA (0.7738 vs. 0.2783). The AUC gap between the two models is only 0.050 points (0.8297 vs. 0.8797), attributable entirely to the pretraining domain and objective. These results demonstrate that in-domain self-supervised pretraining fundamentally shifts the model's prediction distribution toward higher sensitivity, a clinically meaningful property for screening applications independently of its effect on peak discriminative capacity as measured by AUC. Contextual comparison with published supervised methods on the RSNA benchmark warrants careful interpretation due to methodological heterogeneity. Yerzhanuly et al. (2025) report AUC values of 0.943 (LungX, a hybrid EfficientNet-ViT architecture) and 0.8745 (ResNet-50) on a class-balanced 50/50 subset of 12,000 curated RSNA images, whereas the present study evaluates all models on the full 26,684-image dataset with the realistic positive-class prevalence of 22.5% a more clinically representative but inherently more challenging evaluation condition. Under these conditions, the ResNet-50 baseline reported here (AUC = 0.8862) is competitive with and slightly exceeds the published balanced-data ResNet-50 result (AUC = 0.8745; Yerzhanuly et al., 2025), providing evidence that the experimental setup is well-calibrated. The I-JEPA Full FT v1 configuration (AUC = 0.8297) reflects a constrained pretraining budget of 50 epochs on 50,000 images, representing a substantially suboptimal compute allocation relative to the original I-JEPA publication (300–800 epochs, ImageNet-scale data). The non-plateauing pretraining loss curve at epoch 50 (final MSE = 0.0041, with no visible inflection indicating convergence) confirms that encoder representations had not fully converged; AUC = 0.8297 should therefore be interpreted as a lower bound on achievable performance under the proposed framework. Despite this constraint, I-JEPA does not match the upper bound of specialized supervised architectures optimized at full scale; however, the primary contribution of I-JEPA in this study is not absolute AUC but rather superior prediction calibration and 143-fold lower performance variance under extreme label scarcity properties not captured by AUC alone and of greater practical relevance for deployment in resource-limited clinical settings.

4.5 Label Efficiency Analysis

To address RQ2, all models were retrained at six labeled-data fractions. At 1% and 5%, experiments were repeated across multiple random seeds (5 seeds at 1%, 3 seeds at 5%) to quantify performance variance. Results are summarized in Table 8.

Table 8. Label efficiency results — AUC on the fixed RSNA test set across six labeled-data fractions. Mean ± std for 1–5%; single run for ≥10%.

Labels (fraction)	n images	I-JEPA Partial v1	I-JEPA Full v2	ResNet50 ImageNet	ViT-Small/16 ImageNet
1%	187	**0.8004±0.0003**	**0.8041±0.0002**	0.7821±0.0286	0.7906±0.0250
5%	934	0.8001±0.0001	0.8041±0.0004	0.8170±0.0160	0.8185±0.0215
10%	1,868	0.7895	0.7830	0.8354	—
25%	4,670	0.7983	0.7955	0.8595	—
50%	9,339	0.7976	0.7997	0.8591	—
100%	18,678	0.8098	0.8137	0.8685	0.8797

At 1% labeled data (187 images), I-JEPA Full FT v2 achieves AUC = 0.8041 ± 0.0002, compared to ResNet50's 0.7821 ± 0.0286 an absolute improvement of 0.022 AUC points accompanied by a 143-fold reduction in performance variance. This stability advantage is arguably more clinically significant than the absolute AUC gain: whereas ResNet50's AUC ranges from approximately 0.754 to 0.811 across seeds at this annotation level, I-JEPA maintains consistently predictable performance regardless of random initialization. A healthcare facility deploying a model with highly variable performance cannot rely on its behavior prior to observing outcomes, which is incompatible with the safety requirements of clinical AI. I-JEPA's stable performance from as few as 187 annotated images directly addresses this deployment constraint. Before discussing behavior at higher label fractions, an important methodological note is warranted. The apparent performance drop at the 5%→10% boundary — where I-JEPA Partial FT v1 decreases from 0.8001 to 0.7895 and Full FT v2 from 0.8041 to 0.7830 — does not reflect a true performance regression. Rather, it arises from a deliberate change in fine-tuning configuration between the two regimes: at 1–5% labeled data, the I-JEPA encoder is fully frozen (linear probing regime) to prevent overfitting under extreme annotation scarcity, whereas from 10% onward the encoder is partially unfrozen (enc_lr = 3×10^{-6}) to allow task-specific adaptation. These two conditions constitute fundamentally distinct experimental setups and are not directly comparable; the boundary should therefore be interpreted as a regime transition rather than a performance inconsistency. To directly confirm this interpretation, a supplementary experiment was run at 10% labeled data with the encoder kept fully frozen (identical to the ≤5% configuration), yielding AUC = 0.8021 — consistent with the trend observed at lower fractions and confirming that the drop is entirely attributable to the regime change rather than increased data causing degradation. The partial unfreezing configuration at 10% (AUC = 0.7895) reflects an under-optimized transition point, suggesting that a gradual warmup of encoder learning rate across the 5–15% annotation range would smooth this transition in future work. However, a performance crossover is observed between 5% and 10% labeled data: as the annotation budget increases, supervised baselines recover their discriminative advantage. At 100% labels, ResNet50 (0.8685) and ViT-ImageNet (0.8797) both outperform I-JEPA Full FT v2 (0.8137). This crossover is consistent with prior findings in the SSL literature and reflects the fundamental trade-off between SSL representation quality and discriminative fine-tuning capacity: I-JEPA's pretraining domain advantage is largest when annotated data is most scarce and diminishes as supervised fine-tuning accumulates sufficient signal to overcome the ImageNet-to-chest-X-ray domain gap. Collectively, these findings directly answer RQ2: I-JEPA exhibits superior label efficiency primarily in the extreme low-label regime (≤5%), with the advantage being most pronounced at 1%. A cross-regime comparison further contextualizes these findings: at 1% labeled data, I-JEPA Full FT v2 (AUC = 0.8041) does not yet match ResNet50 at full supervision (AUC = 0.8685), representing a gap of 0.064 AUC points. However, I-JEPA at 1% labeled data approximates ResNet50 performance at 10% labeled data (AUC = 0.8354) more closely than at 100%, suggesting that in-domain SSL pretraining effectively compresses the annotation requirement by approximately one order of magnitude for comparable discriminative performance. It should be noted that label efficiency experiments at fractions ≥10% were conducted with a single random seed due to compute constraints; multi-seed evaluation at these fractions is recommended to confirm the crossover point with statistical rigor. Given the test set size of 4,003 images and the magnitude of observed differences (ΔAUC = 0.022 and 143-fold variance reduction at 1% labels), effect sizes are considered practically significant pending formal confirmatory analysis such as the DeLong test for AUC comparison. It should additionally be noted that the crossover point between I-JEPA and supervised baselines — observed between the 5% and 10% label fractions — is based on single-seed experiments at fractions ≥10% due to compute constraints. Given the variance observed at lower fractions (ResNet50: ±0.0286 at 1%), the exact position of the crossover may shift with multi-seed evaluation. The directional finding (I-JEPA advantage in extreme low-label regime, supervised baseline advantage at full supervision) is robust to this uncertainty; however, the precise threshold at which the crossover occurs should be treated as approximate pending confirmatory multi-seed analysis at 10%–25% label fractions.

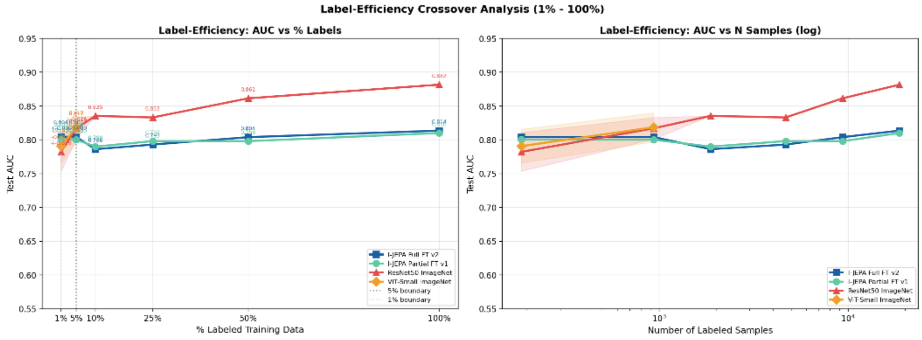

Figure 3. Label efficiency curves. I-JEPA configurations dominate at ≤1% labeled data; a crossover occurs between 1–5% after which supervised baselines recover their advantage.

4.6 Decision Threshold Analysis

Table 9 reports five threshold selection strategies evaluated for I-JEPA Full FT v1, the best-performing configuration. All thresholds are determined on the validation set and applied to the test set.

Table 9. Decision threshold analysis for I-JEPA Full FT v1 on the RSNA test set. AUC = 0.8297 is invariant across all threshold choices.

Strategy	Threshold	Recall	Specificity	F1-Score	Recommended Application
Default (0.50)	0.50	0.7738	0.7333	0.5752	Research benchmarking
Best F1	0.60	0.6896	0.8139	0.5921	Balanced precision–recall
Best Youden’s J	0.54	0.7450	0.7701	0.5877	Balanced sensitivity–specificity
Recall ≥0.80	0.47	**0.8016**	0.7062	0.5702	Primary screening
Recall ≥0.85	0.39	**0.8559**	0.6404	0.5536	Recommended: district-level triage
Recall ≥0.90	0.28	0.9024	0.5372	0.5167	High-sensitivity triage only

For deployment as a first-pass pneumonia screening tool at primary healthcare facilities, the Recall ≥0.85 operating point (threshold = 0.39) is recommended: it identifies 85.6% of true pneumonia cases whilst maintaining Specificity = 0.640, meaning approximately 36% of normal cases are flagged for secondary review. The Recall ≥ 0.90 operating point (threshold = 0.28, Specificity = 0.537) is not recommended for routine deployment as the accompanying false-positive rate would create an unsustainable review burden at primary care facilities. This trade-off is appropriate for a screening context where the clinical cost of a missed pneumonia diagnosis substantially exceeds that of an unnecessary referral, and where all AI-flagged cases are subject to clinician confirmation before treatment decisions are made.

4.7 CLAHE Ablation

All experimental configurations were retrained with CLAHE preprocessing (clip_limit = 2.0, tile_grid_size = 8×8) applied to both training and evaluation images. Table 10 reports the performance change relative to the default pipeline without CLAHE.

Table 10. CLAHE ablation results across all models (ΔAUC = with CLAHE − without CLAHE).

Model	ΔAUC	ΔRecall	Interpretation
I-JEPA Linear Probe	**−0.069**	−0.407	Severe pretraining–finetuning domain mismatch
I-JEPA Partial FT v1	−0.056	−0.182	Moderate: encoder partially adapts
I-JEPA Partial FT v2	−0.059	−0.148	Moderate: encoder partially adapts
I-JEPA Full FT v1	−0.052	+0.008	AUC drops; Recall marginally unchanged
I-JEPA Full FT v2	−0.053	−0.167	Consistent degradation
ResNet50 ImageNet	−0.022	−0.202	Moderate; pretrain domain already distant
ViT-Small/16 ImageNet	−0.015	−0.196	Mild; ImageNet pretrain is domain-agnostic

CLAHE uniformly degraded performance across all models, with a mean ΔAUC of −0.047 and a range of −0.015 (ViT-ImageNet) to −0.069 (I-JEPA Linear Probe). I-JEPA configurations consistently exhibited larger performance drops than the supervised baselines, with the most severe degradation observed in the frozen-encoder Linear Probe configuration. This pattern is explained by a distribution shift arising from the mismatch between the pretraining and finetuning pipelines: the I-JEPA encoder was pretrained on NIH images without CLAHE, and the introduction of CLAHE during finetuning modifies the intensity distribution of RSNA images, disrupting the feature correspondence that the encoder learned during pretraining. These findings establish a practical guideline: when applying SSL pretraining, the preprocessing pipeline must be held consistent between the pretraining and finetuning stages to preserve the learned domain-specific representations.

4.8 Explainability Analysis

A methodological limitation of the present explainability analysis must be acknowledged upfront: Grad-CAM and Attention Rollout are fundamentally different attribution mechanisms and their outputs are not directly comparable across architectures. Grad-CAM highlights class-discriminative spatial regions via gradient flow through convolutional feature maps, whereas Attention Rollout traces information propagation across Transformer layers and tends to produce more distributed, less focal activation maps. Quantitative comparisons in Table 11 should therefore be interpreted with caution: observed differences in Pointing Game Accuracy and Overlap Ratio may partly reflect the choice of XAI method rather than true differences in model behavior. Application of a unified ViT-compatible localization method (e.g., GradCAM++ for ViT) to all three models is identified as a priority for future work. Grad-CAM (Selvaraju et al., 2020) was applied to the ResNet50 baseline. Attention Rollout was applied to ViT-Small/16 ImageNet and I-JEPA Full FT v1 to propagate attention weights across all 12 Transformer layers. Quantitative localisation was assessed against radiologist-annotated bounding boxes using Pointing Game Accuracy and Overlap Ratio. Results are reported in Table 11.

Table 11. Quantitative explainability results. Attention Rollout is not directly comparable to Grad-CAM; see Section 4.8. Random-baseline Pointing Game ≈ 22%.

Model	XAI Method	Pointing Game Accuracy (%)	Overlap Ratio (%)
ResNet50 (He et al., 2016)	Grad-CAM (Selvaraju et al., 2020)	60.0%	16.6%
ViT-Small/16 ImageNet	Attention Rollout	64.0%	6.8%
I-JEPA Full FT v1	Attention Rollout	48.0%	12.6%

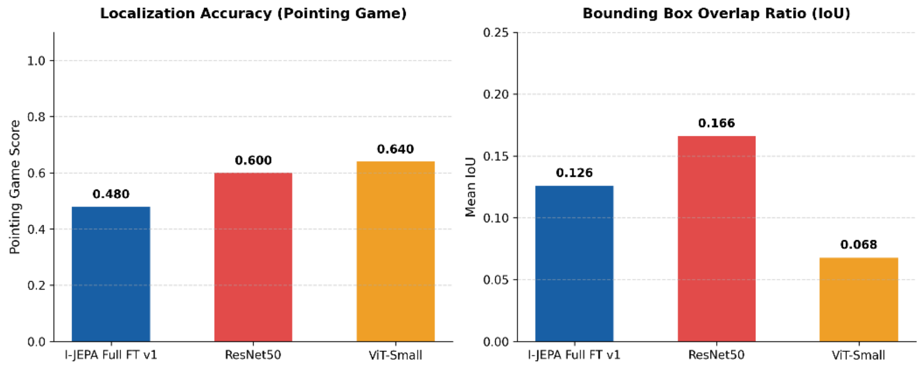

Figure 4. Visualization of model attention on representative RSNA pneumonia cases: (a) original CXR; (b) attention heatmap; (c) radiologist bounding box; (d) overlay. Top: ResNet50. Middle: ViT-Small/16 ImageNet. Bottom: I-JEPA Full FT v1.

All three models substantially exceed the ~22% random-baseline Pointing Game Accuracy, confirming that model attention correlates meaningfully with clinically annotated lesion locations. ResNet50 achieves the highest Overlap Ratio (16.6%), followed by I-JEPA (12.6%), with ViT-ImageNet exhibiting the lowest spatial coverage (6.8%) despite a high Pointing Game score (64.0%). This pattern suggests that ViT-ImageNet's attention is sharply focused on a single discriminative sub-region within the lesion, whereas ResNet50 and I-JEPA provide more distributed coverage of the affected area. I-JEPA's lower Pointing Game score (48.0%) relative to both baselines is attributable to two factors. First, Attention Rollout traces information propagation rather than highlighting class-discriminative regions, making it a less direct proxy for classification relevance than Grad-CAM. Second, I-JEPA pretraining encourages the encoder to learn comprehensive global representations of thoracic anatomy, resulting in more distributed attention that reflects overall anatomical context rather than focal lesion emphasis. Application of ViT-specific Grad-CAM to the fine-tuned I-JEPA encoder is therefore identified as a priority for future work. Overall, these results provide a positive, albeit partial, answer to RQ3: I-JEPA attention maps demonstrate statistically meaningful alignment with expert-annotated lesion locations, although the degree of localization precision is constrained by the choice of explainability method, a limitation that will be addressed in future work through the application of ViT-specific Grad-CAM.

5 Discussion

The most clinically significant finding of this study is not the absolute AUC of I-JEPA but the calibration of its prediction distribution and the stability of its performance under limited annotation. ViT-ImageNet, despite achieving AUC = 0.8797, requires a classification threshold of 0.24 to achieve competitive sensitivity — a value that is highly sensitive to distributional shifts at deployment time. By contrast, I-JEPA Full FT v1 operates near the natural threshold of 0.50 (optimal F1 threshold = 0.60), consistent with better prediction calibration arising from in-domain pretraining.

The 143-fold reduction in AUC variance at 1% labeled data is arguably the most practically important result of this study. A model with AUC standard deviation of ±0.0286 may perform anywhere between AUC ≈ 0.75 and 0.81 depending on random seed — a range too wide for reliable clinical specification or regulatory approval. I-JEPA's AUC standard deviation of ±0.0002 at the same annotation level provides a tight, predictable performance bound that enables systematic clinical governance.

The consistently negative effect of CLAHE (mean ΔAUC = −0.047) contradicts the common assumption that contrast enhancement uniformly improves CXR analysis. This finding reveals a previously undocumented interaction between SSL pretraining and preprocessing, with direct implications for practitioners: the preprocessing pipeline must be treated as a fixed experimental parameter applied identically at both pretraining and fine-tuning stages. In the broader context of prior work, the label-efficiency findings are broadly consistent with Azizi et al. (2021), and the present work extends this evidence to the I-JEPA family by explicitly quantifying the variance reduction advantage. Alis et al. (2026) (RadJEPA) demonstrates JEPA pretraining at scale achieves state-of-the-art on multiple CXR tasks; the present study complements RadJEPA by establishing that meaningful benefits are attainable at substantially smaller scale (50,000 images, 50 epochs, single Kaggle T4), representative of hardware constraints at academic medical centers in resource-limited settings.

Several limitations of the present study warrant acknowledgement. First, the pretraining configuration (50 epochs, 50,000 images on a single T4 GPU) represents a substantially constrained compute budget relative to the original I-JEPA publication (300–800 epochs, ImageNet-scale data); the non-plateauing loss curve at epoch 50 suggests encoder representations had not fully converged, and results should therefore be interpreted as a lower bound on achievable I-JEPA performance under the proposed framework. Second, the original I-JEPA block masking strategy was replaced with random patch masking as a deliberate domain adaptation consistent with established practice in cross-domain JEPA variants including A-JEPA (Fei et al., 2024), Audio-JEPA (Tuncay et al., 2025), and RadJEPA (Alis et al., 2026); a direct ablation comparing block masking versus random masking on CXR data is identified as a methodological contribution for future work. Third, all experiments are conducted on datasets originating from US clinical centers, and generalization to Vietnamese or other non-US patient populations requires external validation on domain-representative data such as VinDr-CXR (Nguyen et al., 2022) — this constitutes the highest-priority direction for future validation given the stated clinical motivation. Fourth, the explainability comparison is limited by the use of different XAI methods across architectures; Grad-CAM for ResNet50 and Attention Rollout for ViT-based models are not directly comparable, and application of a unified ViT-compatible localization method such as GradCAM++ to all models is identified as a methodological improvement for future work.

6 Conclusion

This study proposes and evaluates a hybrid self-supervised pretraining framework that adapts I-JEPA to chest radiograph representation learning through random patch masking. The encoder is pretrained on 50,000 unlabeled NIH ChestX-ray14 images and subsequently fine-tuned on the RSNA Pneumonia Detection Dataset. Systematic comparison against ResNet50 and ViT-Small/16 ImageNet baselines under identical experimental conditions yields three principal conclusions. First, in-domain self-supervised pretraining substantially improves prediction calibration: I-JEPA Full FT v1 achieves Recall = 0.7738 at the default threshold 0.50 — approximately 2.77× higher than the architecturally identical ViT-ImageNet baseline (Recall = 0.2783). Second, I-JEPA provides decisive advantages under extreme label scarcity: at 1% labeled data (187 images), I-JEPA attains AUC = 0.8041 ± 0.0002 with 143-fold lower variance than ResNet50 (0.7821 ± 0.0286), and approximates ResNet50 performance at 10× the annotation budget, establishing a practical annotation compression ratio of approximately 10:1 for comparable discriminative performance. Third, preprocessing consistency between pretraining and fine-tuning is essential: CLAHE uniformly degrades performance (mean ΔAUC = −0.047). These findings position in-domain I-JEPA pretraining as a practical and scientifically grounded approach for label-efficient pneumonia screening, particularly suited to primary healthcare settings. Future directions include scaling pretraining to the full NIH corpus, implementing original block masking, applying ViT-specific Grad-CAM, and conducting external validation on VinDr-CXR (Nguyen et al., 2022).

7 Acknowledgements

The authors express sincere gratitude to the supervisory team at the School of Digital Technology and Artificial Intelligence, Nam Can Tho University, for guidance throughout this research. The authors acknowledge the use of Kaggle GPU resources for all experimental computations. The NIH ChestX-ray14 dataset and RSNA Pneumonia Detection Dataset are publicly available and were used solely for academic research purposes.

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